

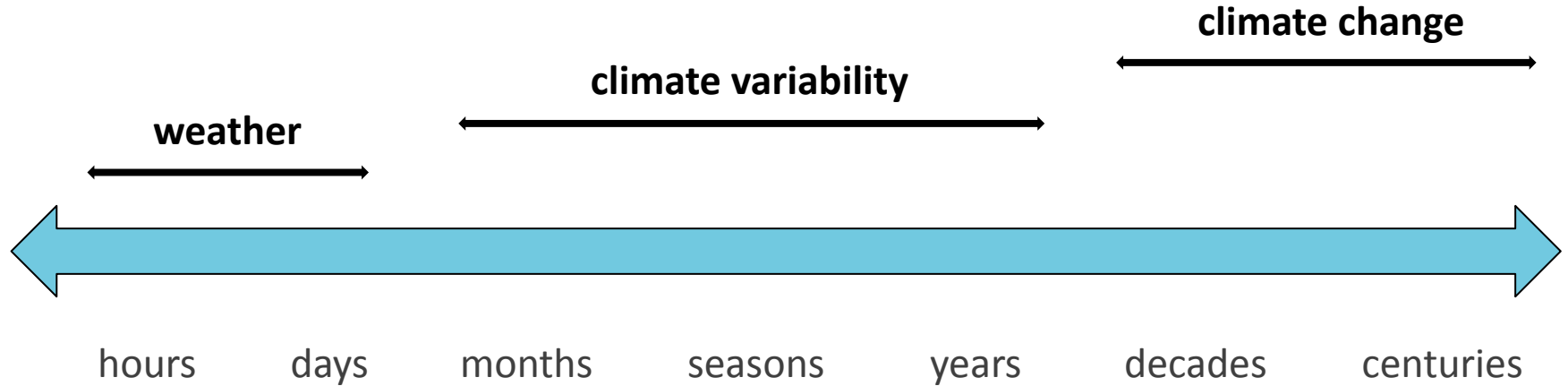
A satellite image of the Alps region, showing snow-capped mountain ranges and surrounding urban areas. Labels for various cities are visible: Dijon, Besançon, Zürich, Bern, Geneva, Lyon, Aosta, Milan, Trento, Venice, Trieste, Ljubljana, Zagreb, Bologna, San Marino, and Florence. The title 'Decoding the climate' is overlaid in large white font with a horizontal line underneath.

Decoding the climate

Essential tips to understanding and analyzing climate data

Alps 2025 - EU, Copernicus Sentinel-3 imagery

Weather vs climate

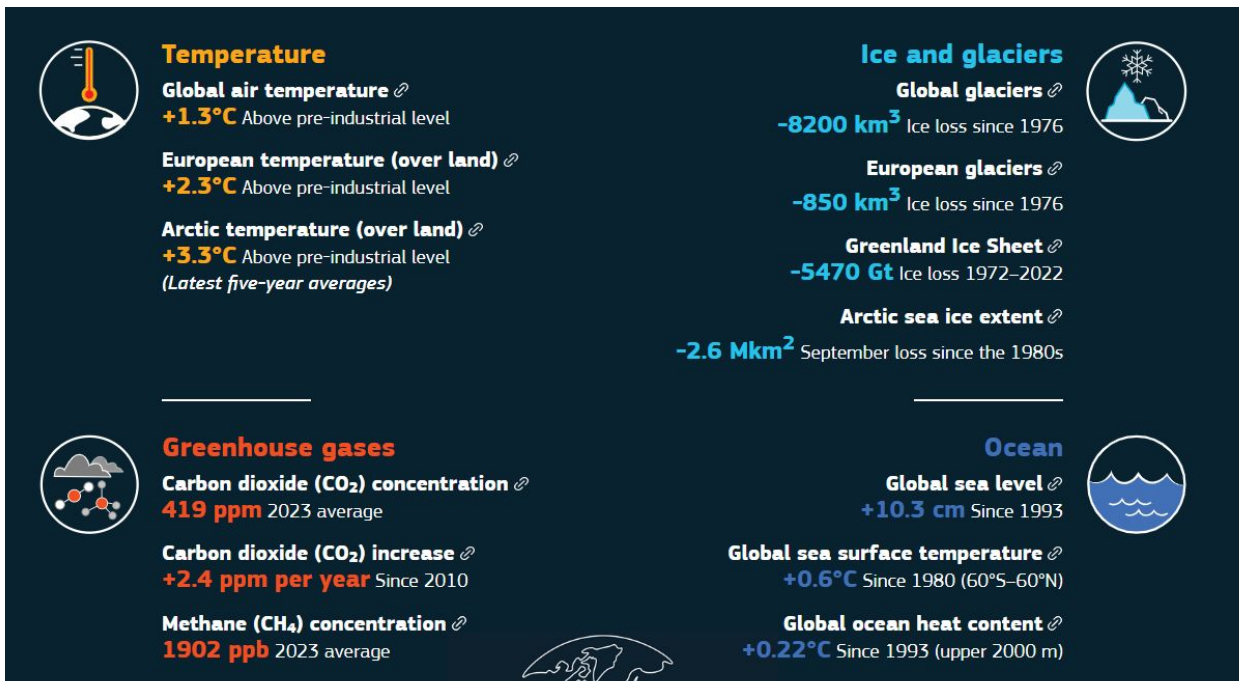


Essential Climate Variables (ECVs)

Atmosphere	Land	Ocean
Surface	Hydrology	Physical
 Precipitation	 Groundwater	 Ocean Surface Heat Flux
 Surface Pressure	 Lakes	 Sea Ice
 Surface Radiation Budget	 River Discharge	 Sea Level
 Surface Temperature	 Terrestrial Water Storage (TWS)	 Sea State
 Surface Water Vapour	 Evaporation from Land	 Surface Currents
 Surface Wind Speed and Direction	 Soil Moisture	 Sea Surface Salinity
 Upper-air Temperature	Cryosphere	 Surface Stress
Upper Atmosphere	 Glaciers	 Sea Surface Temperature
 Earth Radiation Budget	 Ice Sheets and Ice Shelves	 Subsurface Currents

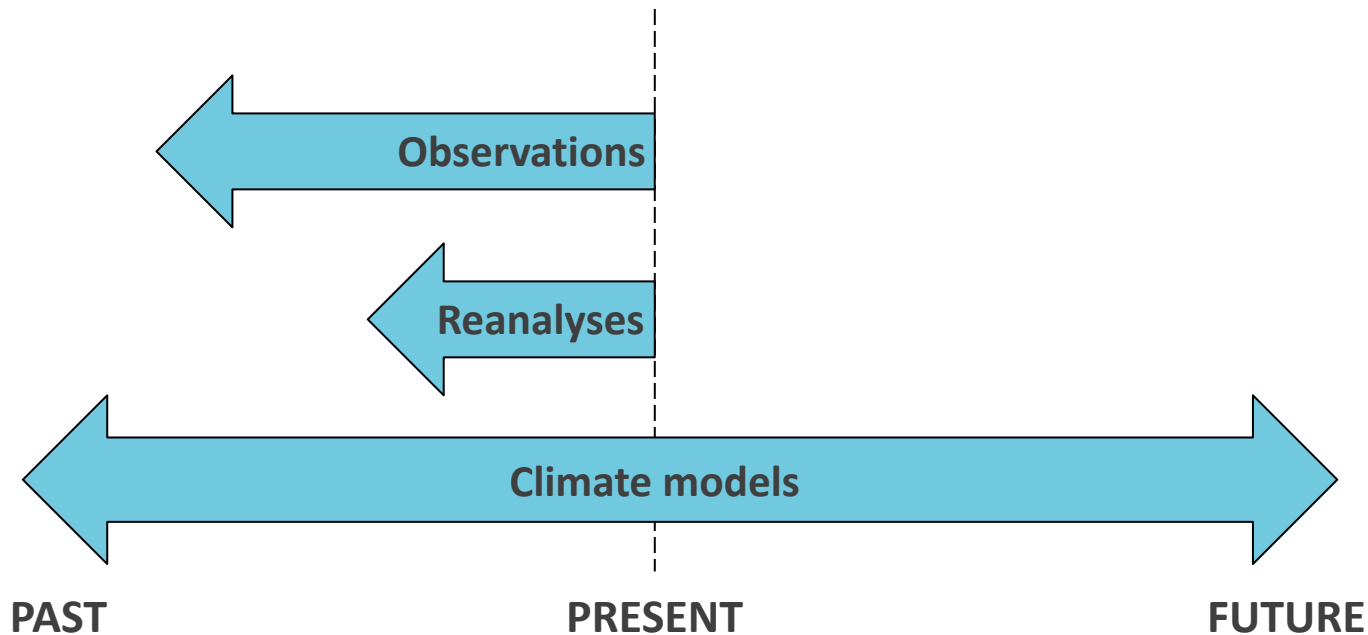
[Global Climate Observing System \(GCOS\) website](#)

Climate indicators



[Copernicus website](#)

Data sources for climate variables



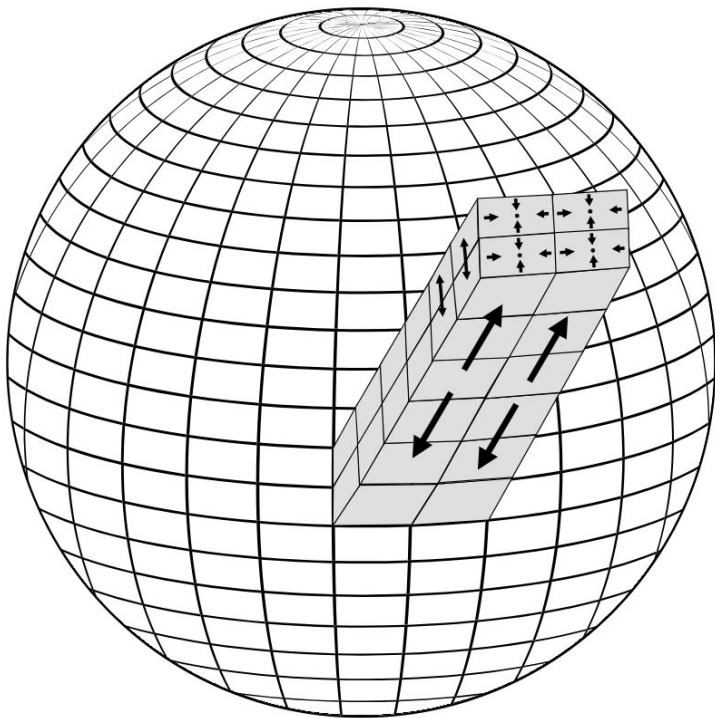
Climate models

HORIZONTAL GRID

latitude - longitude

VERTICAL GRID

height or pressure



HORIZONTAL EXCHANGE

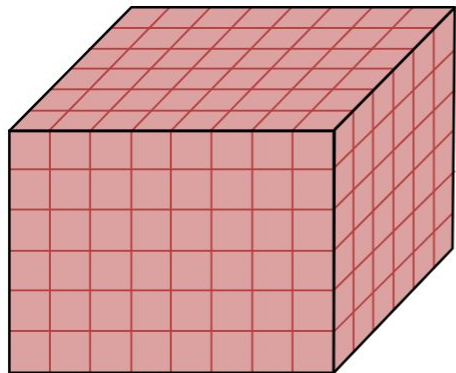
between columns

VERTICAL EXCHANGE

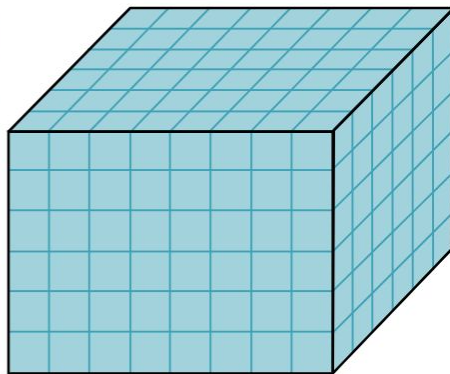
between layers

Climate models

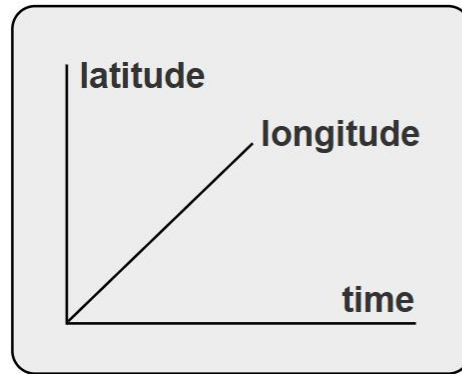
temperature



precipitation



Dimensions

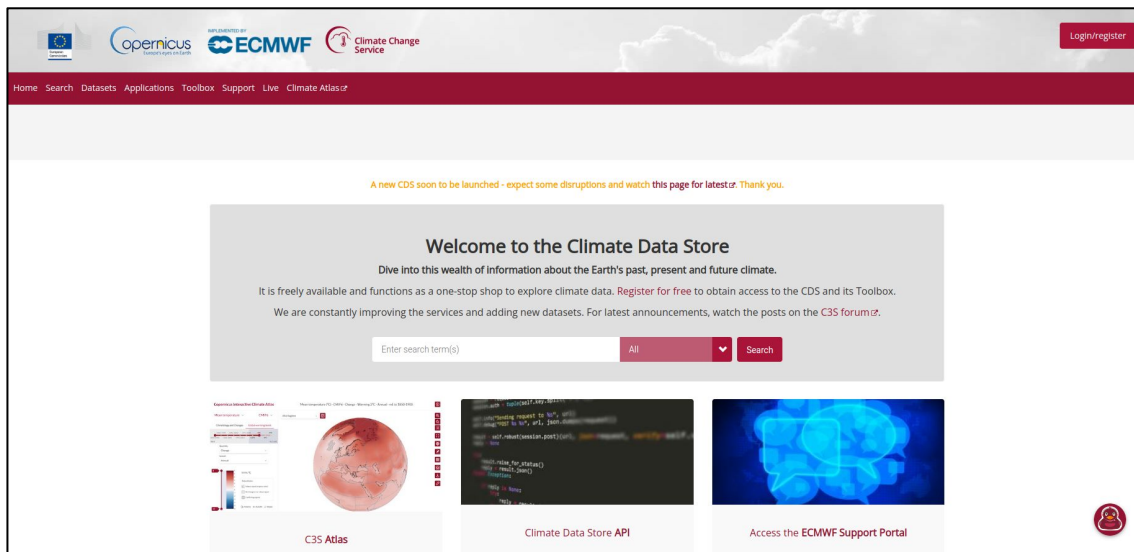


Coordinates

+ Attributes

Where to find data

Copernicus Data Store



<https://cds.climate.copernicus.eu/>

Time to Code

Jupyter notebook

<https://github.com/PyconClimateTrip/pycon2025>

Data

https://drive.google.com/drive/folders/1792N_jbOM6-OwvE8rXaqnb2Nilfu8uKI?usp=sharing